Manoj Kumar Sahukar

Cybersecurity & Observability Architect | Privacy & Data Governance | Cloud Security | ISC2 CC

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Profiles

LinkedIn
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Summary

Cybersecurity and Observability Architect and Privacy Champion at Schneider Electric on EcoStruxure Building Activate, a cloud-based building management platform. Focused on security observability, privacy and data governance: leads product security certification (cloud security, application security, final security review) and privacy by design aligned with India's DPDP Act, GDPR and IEC 62443. Building expertise in AI data governance. ISC2 Certified in Cybersecurity (CC). Brings a hands-on IoT, hardware and cloud background built since 2019 across IIT Delhi, Zenatix (now part of Schneider Electric) and Fab Academy.

Experience

Schneider Electric (via Zenatix acquisition) Gurugram, India
Senior Cybersecurity Engineer – EcoStruxure Building Activate Jan 2022 – Present

- Joined Zenatix Solutions Pvt Ltd as Senior R&D Engineer (Jan 2022); continued as the company became Zenatix by Schneider Electric
- Current role: Senior Cybersecurity Engineer, also serving as IT Owner, Privacy Champion and Cybersecurity and Observability Architect
- Work on EcoStruxure Building Activate, Schneider Electric's cloud-based building management platform
- Led change management across the cloud-based Activate solution, rolling out security enhancements and compliance adaptations with minimal disruption to operations
- Led product security certification requirements, including cloud security, application security, final security review and privacy by design
- Applied regulatory and industry security standards to the platform: ISA/IEC 62443 and GDPR, and more recently India's Digital Personal Data Protection (DPDP) Act 2023
- Supported digital integration initiatives spanning DNS, WAF, cloud EDR, proprietary identity and access management (IAM) platform, GlobalProtect VPN, email and GitHub Enterprise

Sahukar Consultancy (Curiosta.com) Odisha, India
Freelance Consultant (part-time) Feb 2019 – Feb 2020

- Freelance consulting and product work through a sole proprietorship, Sahukar Consultancy, which runs Curiosta.com
- Led product and architecture for Curiosta MMS, an open-source makerspace inventory and lending app (makers.curiosta.com): member issue, borrow and return requests; admin approvals, picking and returns; Google Sheets inventory sync
- Re-architected it in 2026 to a Preact PWA with a Medusa (Node.js) backend on AWS Lambda (SAM), Neon Postgres, S3 and CloudFront, running for about US$0.15 a month; serious and critical accessibility issues cut from 195 to 0

makers.curiosta.com

IIT Delhi (IRD) – Dept. of Computer Science & Engineering, Dr. Rijurekha Sen's group New Delhi, India
Project Scientist Dec 2019 – Dec 2021

- DST-funded project on air pollution measurement using sensors mounted on Delhi (DIMTS) buses
- Designed a scalable IoT solution for spatio-temporal measurement of air pollution in Delhi NCR using vehicle-mounted sensors
- Built the hardware units (sensors, microcontrollers) and multiple prototypes; managed deployment in DIMTS buses, cloud data upload and anomaly-detection scripts
- Backend built on Google Cloud Storage, Dataflow, BigQuery and VMs; frontend on ReactJS and RESTful API
- Embedded device developed on STM32 ARM MCUs, SIM7600 and Raspberry Pi; programs in STM32 HAL, ESP32 IDF and Python
- Delhi Pollution Dataset: https://www.cse.iitd.ac.in/pollutiondata

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Aerogram Pvt Ltd — New Delhi, India

Startup CTO — Aug 2019 – Dec 2021

- Responsible for end-to-end product development of air quality monitoring devices
- Developed and deployed a scalable IoT platform (HTTP, MQTT) into production on Google Cloud Platform and Firebase
- Led development of 10+ embedded products (WiFi, 3G, 4G, BLE, NB-IoT) and 5+ data management platforms
- Mentored a team of 10+ engineers across web, CAD, power electronics, embedded and cloud engineering
- DevOps, Scrum Master

Sahukar Ventures (OPC) Pvt. Ltd. — Odisha, India

Founder — Feb 2019 – Feb 2020

- Built a working prototype of a Spidercam-style cable-driven parallel robot for scalable small-farm automation (seeding, watering, monitoring), as the Fab Academy 2019 group project at Vigyan Ashram (gitlab.com/manojsahukar/make-a-machine)
- Derived the cable-length kinematics and 3D force-equilibrium cable tensions for any X, Y, Z position; these drove the control system and stepper motor selection
- Designed and fabricated mechanism parts in Fusion 360 and ran the 9-person team as Scrum Master (weekly sprints, issue board, live wiki)
- Closed the venture after the prototype didn't attract enough customers

GitLab: make-a-machine

Elegate — Cairo, Egypt

Business Developer (AIESEC Global Entrepreneur Internship) — Jul 2018 – Nov 2018

- Responsible for pricing, process automation, technology management, industry analysis, market research and product roadmap
- Developed browser automation scripts in NodeJS

Curiosta Innovation Labs LLP — Durgapur, West Bengal, India

Co-Founder — Jan 2018 – Nov 2019

- Co-founded a makerspace; designed and taught maker courses on Arduino, CAD, 3D printing, rapid prototyping and digital fabrication
- Built an MVP of a makerspace management system for B2B clients
- B2B adoption was too slow to be viable, especially after the government launched its Atal Tinkering Labs, so the venture wound down

IC&SR, IIT Madras — Chennai, India

Project Associate — Nov 2014 – Jul 2016

- Computational modelling and experimental characterization of smart materials and composites
- Co-authored a research paper on PMN-PT ceramics

Tata Consultancy Services Ltd — India

Assistant System Engineer — Jun 2014 – Oct 2014
- Trained in Business Intelligence and Performance Management

Education

National Institute of Technology, Durgapur — M.Tech • CGPA 8.21
Computer Science & Engineering — Durgapur, India • Jul 2016 – Jun 2018
Entrepreneurship & Innovations
Institute Gold Medal (2018)

Siksha 'O' Anusandhan University — B.Tech • CGPA 8.35
Mechanical Engineering — Bhubaneswar, India • Jul 2010 – May 2014
Coordinator of the Cyber Security club

Projects

Hobby websites (curiosta.com subdomains)
Static sites I plan and run with AI agents under my direction, hosted on GitHub Pages with GitHub Discussions comments:

- kinetic.curiosta.com: the physics behind 26 adventure sports
- buildwith.curiosta.com: a source-checked guide to hands-on festivals and contests worldwide
- hawkeye.curiosta.com: a top-down SWOT summary of the world
- founder.curiosta.com: plain-language business strategy for Indian small businesses
- mind.curiosta.com: how to learn and how not to be fooled
- learn.curiosta.com: timeless fundamentals for a world with AGI
- nest.curiosta.com: AI-resilient physical careers and verified Indian job fairs
- equity.curiosta.com: an open notebook on Indian stock picking, results good or bad

kinetic.curiosta.com

Odia Wikipedia volunteer: Odia language tools

- Built open-source converters that turn legacy non-Unicode Odia fonts (OR-TT Sarala, Akruti Sarala, Shreelipi) into Unicode, first in Game Maker Language (released 12 Dec 2012) and then in HTML and JavaScript, so newspapers and books could be reused on Wikipedia and the web
- Built TypeOdia, an offline Odia typing tool adapted from Wikimedia's jQuery.ime, and added the OdiScript input method, deployed on Odia Wikipedia in Feb 2016
- 2016: built a SAMBAD-EML font converter for Sambad (Eastern Media) with fellow volunteers; it was integrated into the CMS of Sambad's new website
- Featured by the Centre for Internet and Society (Dec 2012), Wikimedia's blog (Jun 2014) and an editorial in The Samaja (Jul 2014)

github.com/OdiaWikimedia/converter

Publications

Effect of grain size on piezoelectric, ferroelectric and dielectric properties of PMN-PT ceramics
Ceramics International 45 (5), 5731-5742

Acoustic alliance with bell plate
Advanced Science Letters 20 (3-4), 593-596

Volunteering

Odia Wikipedia community
Contributed to FOSS and Odia Wikipedia science articles

Certifications

Certified in Cybersecurity (CC) — Sep 2025
ISC2

Skills

Observability & Security Operations

Observability architecture, Security monitoring, Anomaly detection, Change management, IT ownership

Cloud & Application Security

Cloud security, Application security, Secure development lifecycle (SDL), Final security review, WAF, DNS security, Cloud EDR, Identity and access management (IAM), VPN (GlobalProtect), Email security, GitHub Enterprise

Privacy, Compliance & Data Governance

Privacy by design, India DPDP Act 2023, GDPR, ISA/IEC 62443, Product security certification, Privacy Champion

Cloud & Architecture

AWS (Lambda/SAM, S3, CloudFront), Google Cloud (BigQuery, Dataflow, Cloud Storage), Firebase, Neon Postgres, Serverless, cost-optimised design, REST APIs, MQTT

Data & Analytics

BigQuery data pipelines, ETL (Informatica), Business Objects, Google Sheets data sync

Development

Python, JavaScript / Node.js, React / Preact (PWA), Medusa, C, C++, C#, Java, Git, CI/CD, GitHub Pages

AI-assisted Engineering

Directing AI agents to plan and run eight websites

Embedded & IoT

STM32 (HAL), ESP32 (IDF), AVR, Raspberry Pi, 4G / NB-IoT / BLE, Field sensor deployment

Hardware & Fabrication

Fusion 360, Cable-driven robot kinematics, 3D printing, Digital fabrication

Leadership

Scrum Master, Mentored 10+ engineers, Taught maker courses

Accessibility

Accessibility audits (195 serious/critical issues to 0)

Languages & Localisation

Odia, Hindi, English, Odia Unicode converters and input methods

Awards

IMPACT Makers Spotlight — Mar 2025
Schneider Electric Digital Buildings, Q1 2025 Open Line

Zenster of the Year 2023 — 2023
Zenatix

Institute Gold Medal, M.Tech (Entrepreneurship & Innovations), NIT Durgapur — 2018
NIT Durgapur